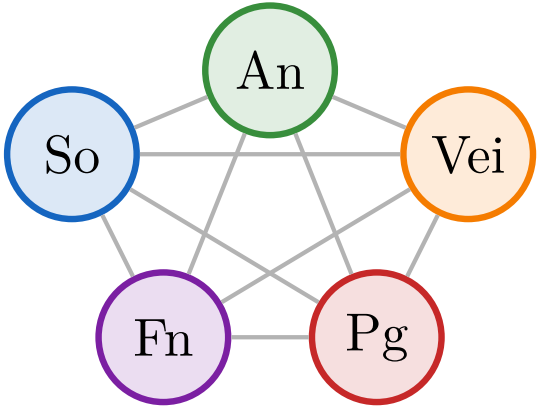
Living material



variational multiphase model
15 unknown interactions a_{ij}

Bayesian calibration

composition

↓

+ viability

TMCMC on a GPU (JAX)
3 seeds per stage,
explicit convergence and
identifiability checks

What the data determine

5–9 of 15
interactions
constrained

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in silico, without
F. nucleatum:
late rise of
P. gingivalis
reduced fourfold